AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 36 · STUDENT LESSON

Expo and Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Weeks 29–35: connect requirements, a decision trace and evaluation evidence.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Evaluate a saved candidate and a manual baseline fairly. Try the recall before opening notes.

Your learning target

I can evaluate a saved candidate and a manual **baseline** fairly.

I can defend a conclusion using results, failures and limits.

I can explain the rules, privacy choices and **human control** in your project.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Understanding is shown by explanation, **transfer**, testing, and limits.



Your target

LEARNING TARGET Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

Learner name:

Readiness for the connected explanation

You change a candidate after seeing a final test fail. Can the same case be an unseen final test of the new version? Explain.

This week's end goal

By the end, I can defend a project using measured results, limits and an honest **baseline** comparison.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Expo and Defense case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A team presents the Classroom Sorter, traces a new item live, shows test **evidence**, explains one failure, and states when a human must decide.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

This week you will evaluate your activity adviser and explain what its **evidence** supports. A project with an honestly reported failure can show strong understanding.

Save your candidate instructions, version label, **input** rules, development records and known problems before your teacher reveals the final requests and catalogue. Keep the instructions unchanged during this evaluation.

The agreed rule remains: validate whole available minutes from 0 to 30 and Quiet required Yes or No. No allows quiet or nonquiet activities.

Start at the first row and suggest the first activity whose time is no more than available time AND whose quiet label satisfies the request. Explain the choice. If all rows fail, report no match.

Invalid or missing **input** needs clarification before searching. Clear the previous **output** for each run.

Your teacher will supply a fresh, labelled catalogue in the same format. Its values are test **input**s, not permission to rewrite your instructions. All materials are assumed available.

The manual **baseline** checks every row against the same requirements, then selects the first fit. Both methods advise only; neither books, publishes or chooses on a person's behalf.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Expo and Defense. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



INPUT

fresh item and documented system



PROCESS

execute and explain each stage under questions



INPUT fresh item and documented system

PROCESS execute and explain each stage under questions

OUTPUT result, **evidence**, **limitation**, and responsible next action

Key idea

CORE IDEA Understanding is shown by explanation, **transfer**, testing, and limits.

How the mechanism works

UNDER THE HOOD A **defense** evaluates **transfer**able understanding, not presentation polish.

Strong **evidence** includes system architecture, version history, planned tests, failures, controlled improvement, fairness/privacy boundaries, and independent response to a novel case.

Draw a fresh example with all three stages:

A closer explanation

From saved version to supported claim

BEFORE THE RUN	DURING THE RUN	AFTER THE RUN
Save instructions and label	Use the assigned data	Keep results and traces
Keep final cases reserved	Run both methods fairly	Compare full passes
State rules and limits	Record actual outputs	Explain failures and limits

Changing instructions during evaluation creates a different version. Keep evidence for the original version.

Use the same final requests, catalogue, requirements and full-pass rule for both methods. Record the candidate **output** and reason before looking at the expected answer.

Record the **baseline** separately; do not copy the candidate result into it. Your teacher holds the answer key.

A full pass needs both the required **output** and a valid reason, including why earlier rows fail when a later row is selected.

For each request record its ID, **input**s, catalogue label, candidate label, candidate **output** and reason, **baseline output** and reason, and both results after checking. Keep a trace for any failure. Mark unrun cases explicitly.

Do not silently correct an **observation**. If a person misreads an instruction, record that execution error separately from a defect in the written instruction.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Expo and Defense. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Polished poster Clue: communication aid. Meaning: not proof alone.

The case to investigate

CASE 2 — Fresh trace Clue: unrehearsed application. Meaning: **transfer evidence**.

The case to investigate

CASE 3 — Failure log Clue: honest testing record. Meaning: engineering **evidence**.

The case to investigate

CASE 4 — Human gate Clue: responsible boundary. Meaning: use condition.

Keep this boundary clear

BOUNDARY Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Someone adds the two unrun cases to the denominator as failures.

Predicted result

Prediction: That reports a different measure, 8/12 completed successes out of planned

cases.

Why the result follows

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Expo and Defense to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
demonstration	live showing of system behavior	
defense	evidence-based explanation under questions	

Word	Meaning	My example
transfer	applying understanding to a fresh case	
limitation	condition where performance or use is restricted	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A successful demo proves every case will work.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Final evaluation	Testing a saved candidate under stated conditions after development
Baseline	A comparison method tested under the same conditions
Evidence	Recorded observations that support or challenge a claim
Defence	An explanation of a decision using evidence and reasoning

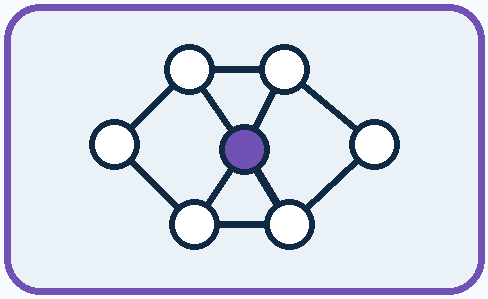
Follow a worked example

> Follow the worked trace for Expo and Defense, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A team presents the Classroom Sorter, traces a new item live, shows test **evidence**, explains one failure, and states when a human must decide.

Step 1

STEP 1 — OBSERVE fresh item and documented system

Step 2

STEP 2 — TRACE execute and explain each stage under questions

Step 3

STEP 3 — CONCLUDE result, **evidence**, **limitation**, and responsible next action

Model answer

MODEL REASONING A complete **defense** traces **input**, **process**, and **output**; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

A claim to avoid

DO NOT OVERCLAIM Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

What new **evidence** could change this answer?

A closer explanation

Worked explanation · Expo and Defense

Here is a fictional example from a different project, not your class results. A candidate has 6 full passes from 8 attempted cases; the **baseline** has 8 from the same 8. Two more planned cases were not run.

The candidate passed 75 percent of attempted cases; the **baseline** passed 100 percent. Neither has **evidence** for the two unrun cases. The candidate did not outperform the **baseline** on this measure.

Evidence	Supported statement	Claim to avoid
6 of 8 full passes	Six attempted cases met the full rule	It works for every request
Both methods pass the same cases	They tie on this measured result	Automation must be better
A time was not measured	Speed is unknown	The candidate is faster
Only fictional catalogue data	Results concern these test conditions	People will enjoy every suggestion

Worked explanation · Expo and Defense

Explain each failure and its consequence instead of hiding it inside a total. Even perfect results on a small reserved set do not establish reliability in every situation.

Our paper adviser applies human-**written rule**s; it has not learned a model from **training** examples. It checks compatibility, not enjoyment, popularity or real material availability.

In the fictional example, state the candidate and baseline results, explain how the two unrun cases should be reported, and give one useful next action. Why is “the adviser is ready for every classroom” unsupported?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Expo and Defense

A fictional paper sorter team prepares an exhibition **evidence** board.

The facts we will use

Supplied facts

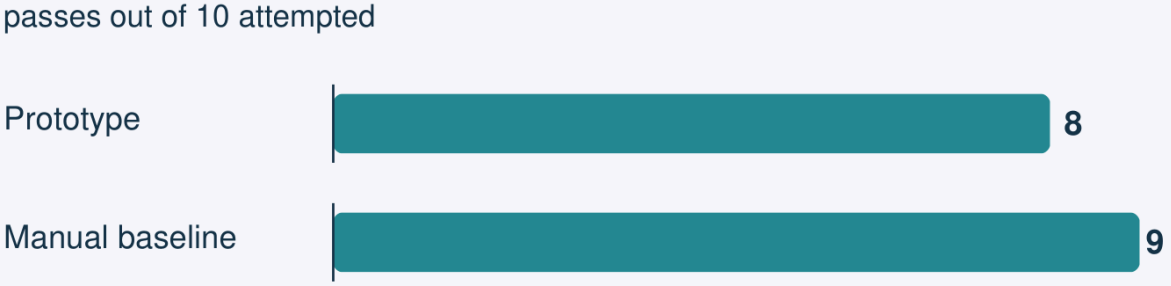
Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Supplied facts

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

AI ACADEMY / AI-1/36

Shared reserved-case comparison



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/36

Expo and Defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Describe contract

Show the task, inputs, rule and reserved case selection.



2. Show results

Prototype 8/10=80%; baseline 9/10=90% on attempted cases.



3. Keep incomplete work

Two unrun cases remain visible, without pass/fail labels.



4. Defend conclusion

Prototype did not beat baseline on this measured outcome; explain its failures.

CHANGE ONE THING

Someone adds the two unrun cases to the denominator as failures.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Describe contract	Show the task, inputs, rule and reserved case selection.
2. Show results	Prototype 8/10=80%; baseline 9/10=90% on attempted cases.
3. Keep incomplete work	Two unrun cases remain visible, without pass/fail labels.
4. Defend conclusion	Prototype did not beat baseline on this measured outcome; explain its failures.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

What this example does not establish

Keep the stated limit

This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Keep the stated limit

Read the labelled picture: Write one defensible success statement and one claim the **evidence** cannot support.

Keep the stated limit

Change one thing: Someone adds the two unrun cases to the denominator as failures. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Expo and Defense claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Polished poster	communication aid	not proof alone
Fresh trace	unrehearsed application	transfer evidence
Failure log	honest testing record	engineering evidence
Human gate	responsible boundary	use condition

Repair two tempting claims

Challenge claim A

CLAIM A A successful demo proves every case will work.

Repair:

Challenge claim B

CLAIM B Admitting a **limitation** lowers the quality of the project.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Archive the evaluated candidate and its original results. If you repair it after evaluation, give it a new version label and use revealed cases as development checks. Do not relabel them as unseen.

Ask your teacher for fresh reserved cases only after the revision is fixed. Do not erase the first result.

Check the claim against the facts

Write one conclusion limited to the actual conditions tested and one specific next step. It may be better to keep the manual **baseline**. Grade 7 will turn carefully stated procedures into programs and examine data more closely.

Carry forward the habits of validating **input**s, recording **evidence** and keeping people in control.

Name one observation from your own evaluation, one limit on your conclusion and one next action. Explain why your adviser is a rule-based program design rather than a trained model, and why passing tests does not give it permission to book an activity.

STUDENT LESSON · 8 OF 18

Practise together

Practise Expo and Defense with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Understanding is shown by explanation, **transfer**, testing, and limits.

Use a fresh example to explain **demonstration**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Admitting a **limitation** lowers the quality of the project.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Prepare a short explanation using your saved instructions, one complete trace, the final comparison, and a **limitation**. State the problem and whose choice the adviser supports.

Explain validation, both matching conditions, first-fit order, reset and stopping. Show the exact instruction behind your trace. Describe any repair with its old **evidence**, or say honestly that no repair was needed.

Use only fictional requests and the two necessary **input** fields. Do not collect names, diagnoses or reasons a pupil prefers quiet. A quiet request is enough.

A person may decline the suggestion; a teacher checks real materials and local rules before an activity. A successful **output** grants no permission to book or publish.

For a two-minute partner presentation, use this order: problem and rule; trace; candidate versus **baseline** results; failure or **limitation**; next action. Partners swap speaker and listener roles.

Each person must explain one trace independently. The listener asks which recorded **observation** supports the main claim; attractive slides cannot substitute for that **observation**.

Practise with less help: Keep the final evaluation record

Use workbook W2 for the connected example “Evaluating and explaining your project”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Write one defensible success statement and one claim the **evidence** cannot support.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

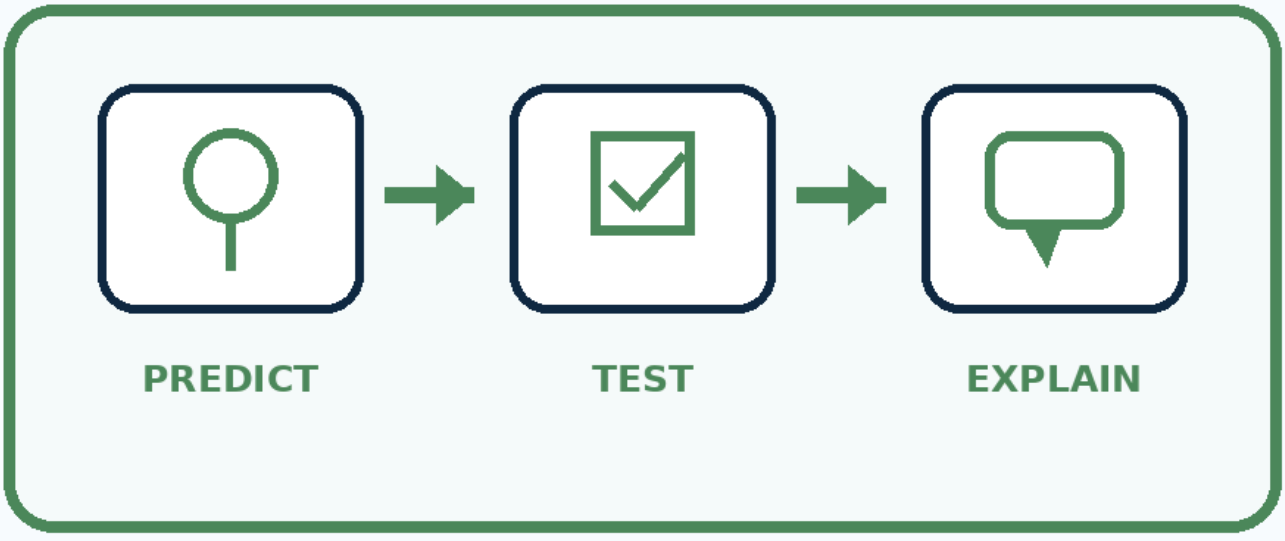
Run the investigation

> Investigate Expo and Defense. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Responsible Classifier Expo

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Demo the system, solve a fresh case, and answer why it may fail.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one **limitation**, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

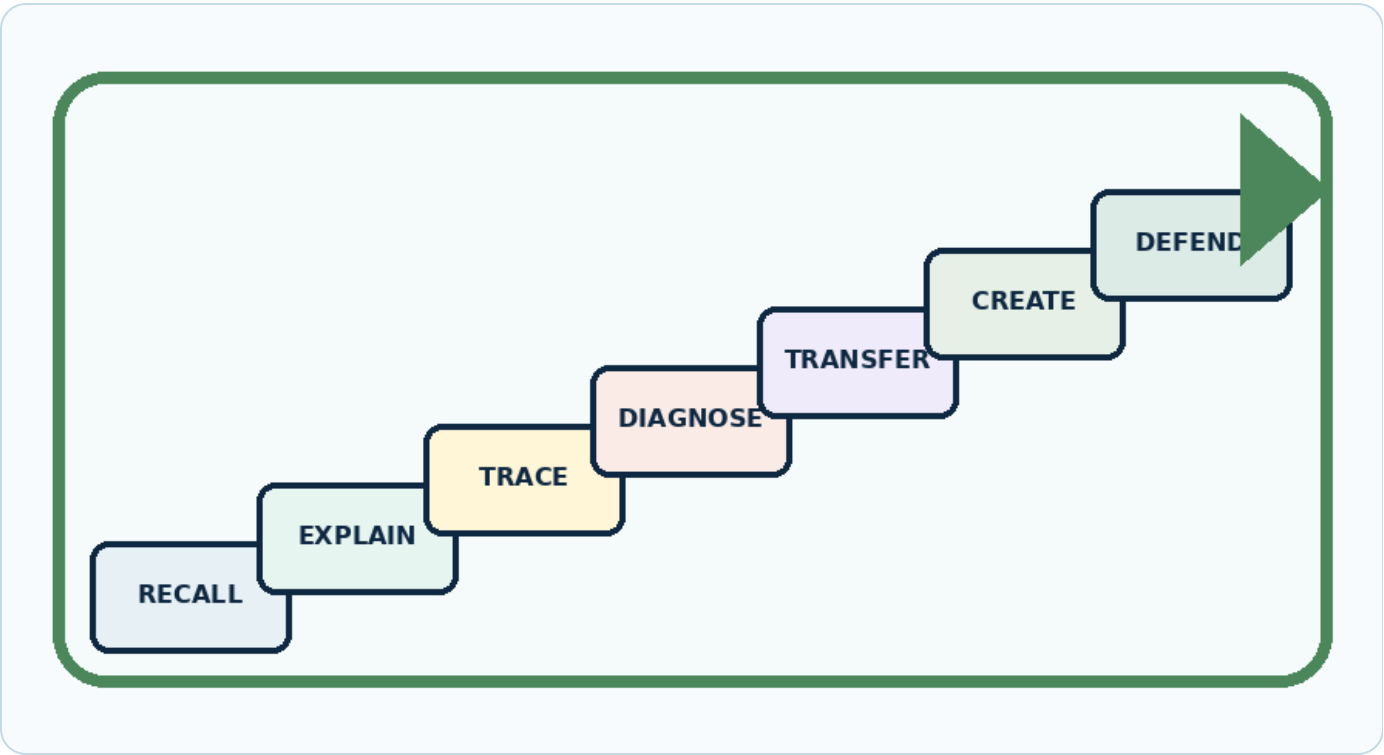
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Expo and Defense. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **demonstration**.

EXPLAIN Understanding is shown by explanation, **transfer**, testing, and limits.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A successful demo proves every case will work.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

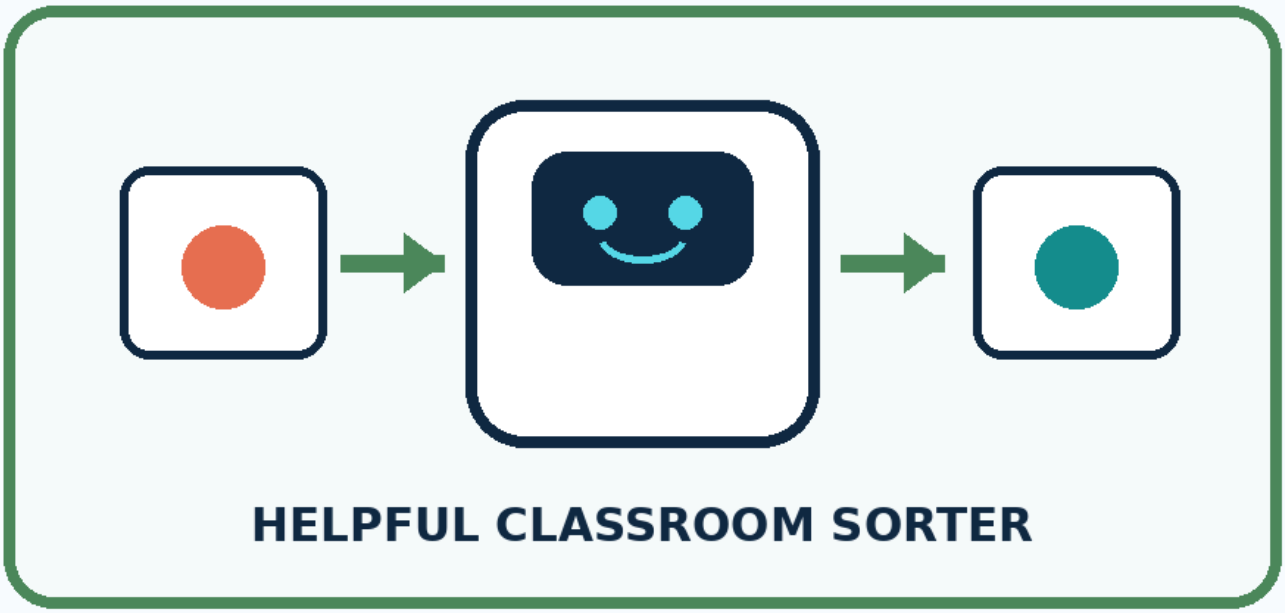
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

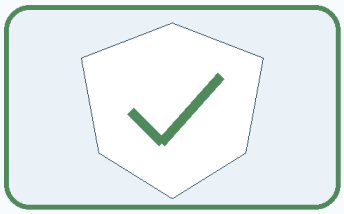
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

My project decision:

Evidence for your project

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Demo the system, solve a fresh case, explain test **evidence**, name limits,

and answer why the system may fail.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK A complete **defense** traces **input**, **process**, and **output**; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: the next Academy level.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Expo and Defense case for supported practice. Repeated examples do not become fresh assessments.

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The expo poster says always correct after six classroom passes.

The question

Write a defensible claim.

Worked explanation

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Defend a claim with a complete trace

Evidence or stage	Value or result
Input	Show a specified test object
Decision	Explain its path

Evidence or stage	Value or result
Output	Show the tray or review result
Evidence	Report six classroom passes and limits

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Judge a different project independently

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Expo and Defense, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional paper sorter team prepares an exhibition **evidence** board.

By the end, I can	How I will show it
Evaluate a saved candidate and a manual baseline fairly	Explain the deciding step, show a trace or test, and state one limit.
Defend a conclusion using results, failures and limits	Explain the deciding step, show a trace or test, and state one limit.
Explain the rules, privacy choices and human control in your project	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Expo and Defense without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Evaluate a saved candidate and a manual **baseline** fairly

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper sorter team prepares an exhibition **evidence** board.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Prototype did not beat **baseline** on this measured outcome; explain its failures.

1. Keep incomplete work

2. Describe contract

3. Show results

4. Defend conclusion

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Defend conclusion Prototype did not beat **baseline** on this measured outcome; explain its failures. An exhibition should make the **evidence** inspectable.

A polished **demonstration** is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster.

Unrun cases must not be treated as successes.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Someone adds the two unrun cases to the denominator as failures. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. That reports a different measure, 8/12 completed successes out of planned cases.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That reports a different measure, 8/12 completed successes out of planned cases.

> It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Repair a missing building block

Review Improve One Cause (ai-1/35). Goal: Repair a located defect without changing the agreed requirements. Short practice: Explain why “our repaired version now picks something” is not enough to show success at 6..

Return to this week and explain what you changed.

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

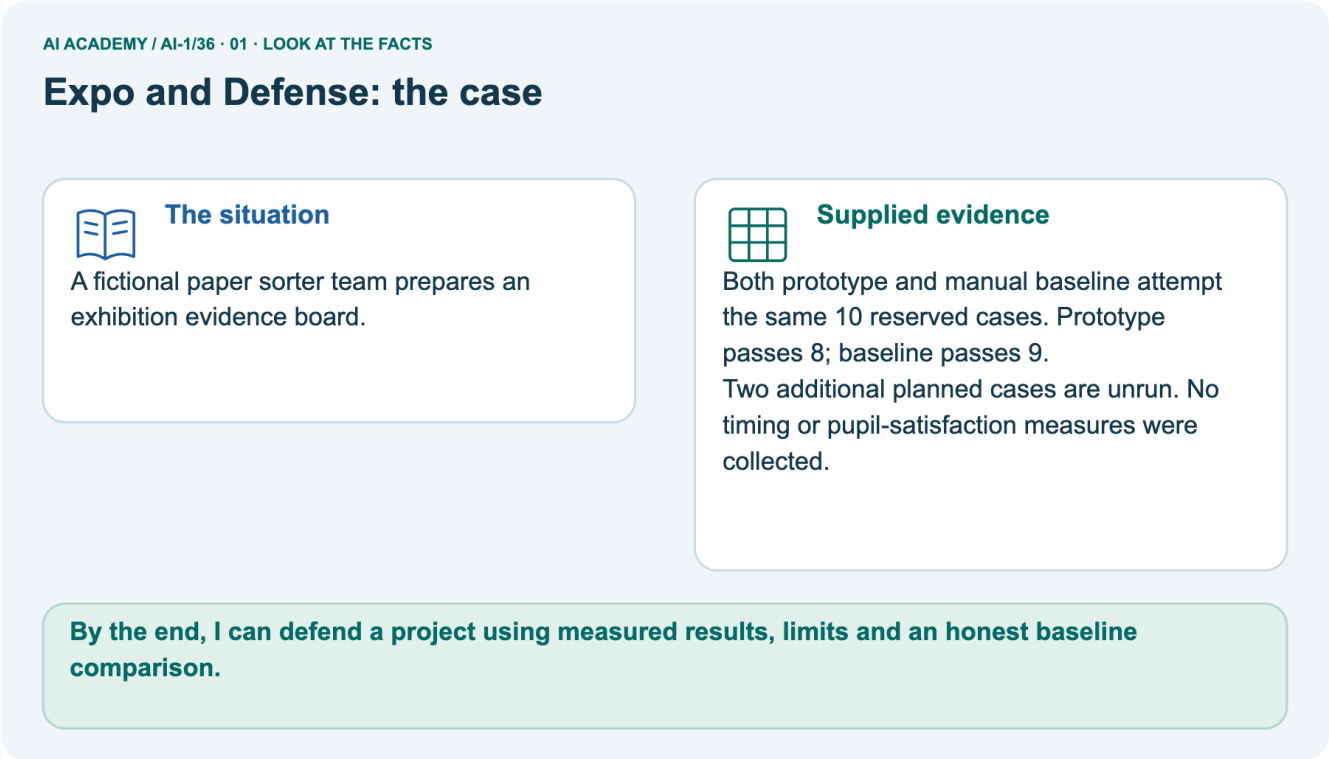
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Expo and Defense graphic using its labels, deciding evidence and relevant limitation.

01. Expo and Defense: the case



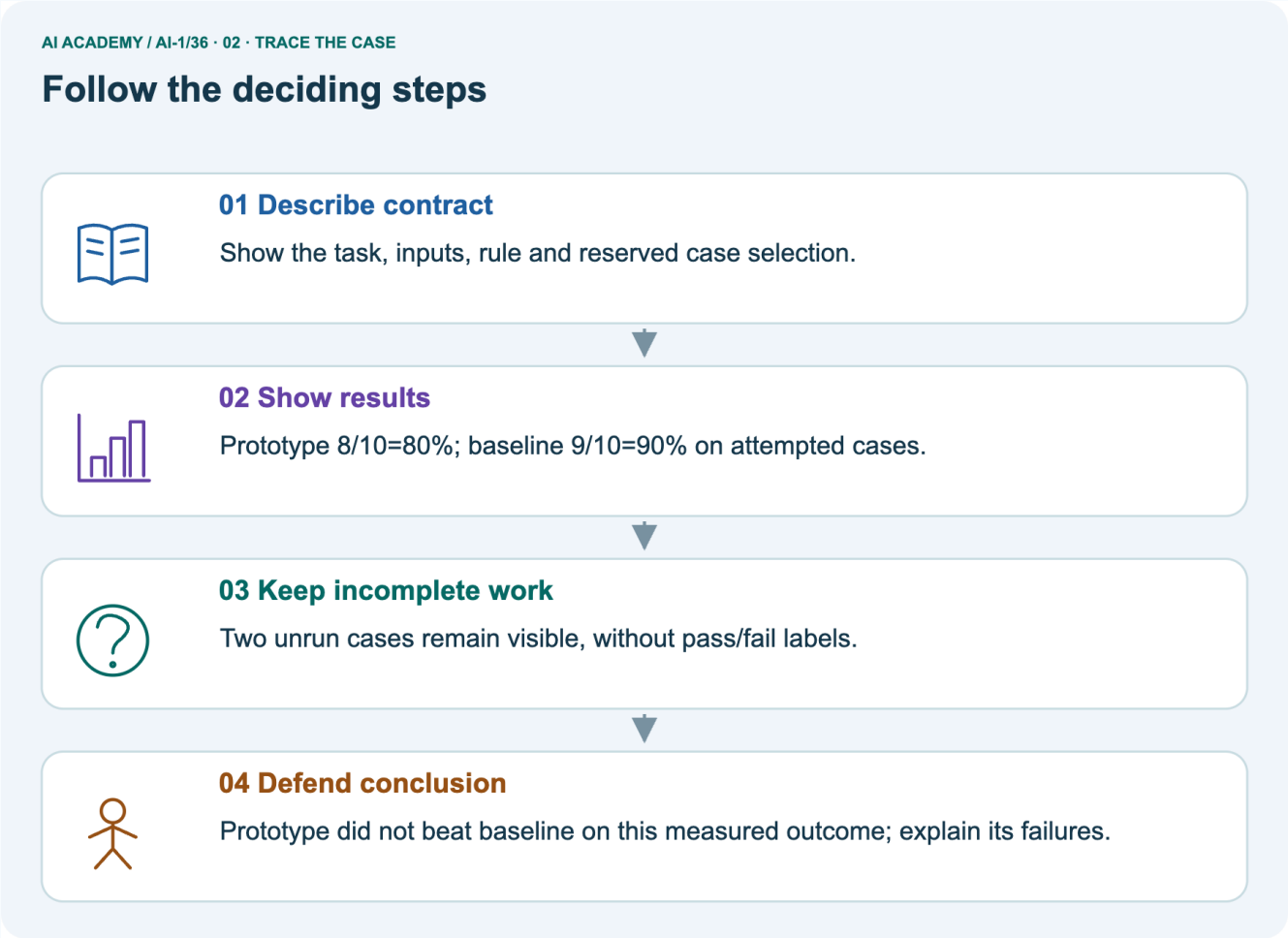
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter team prepares an exhibition **evidence** board. Supplied facts: Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

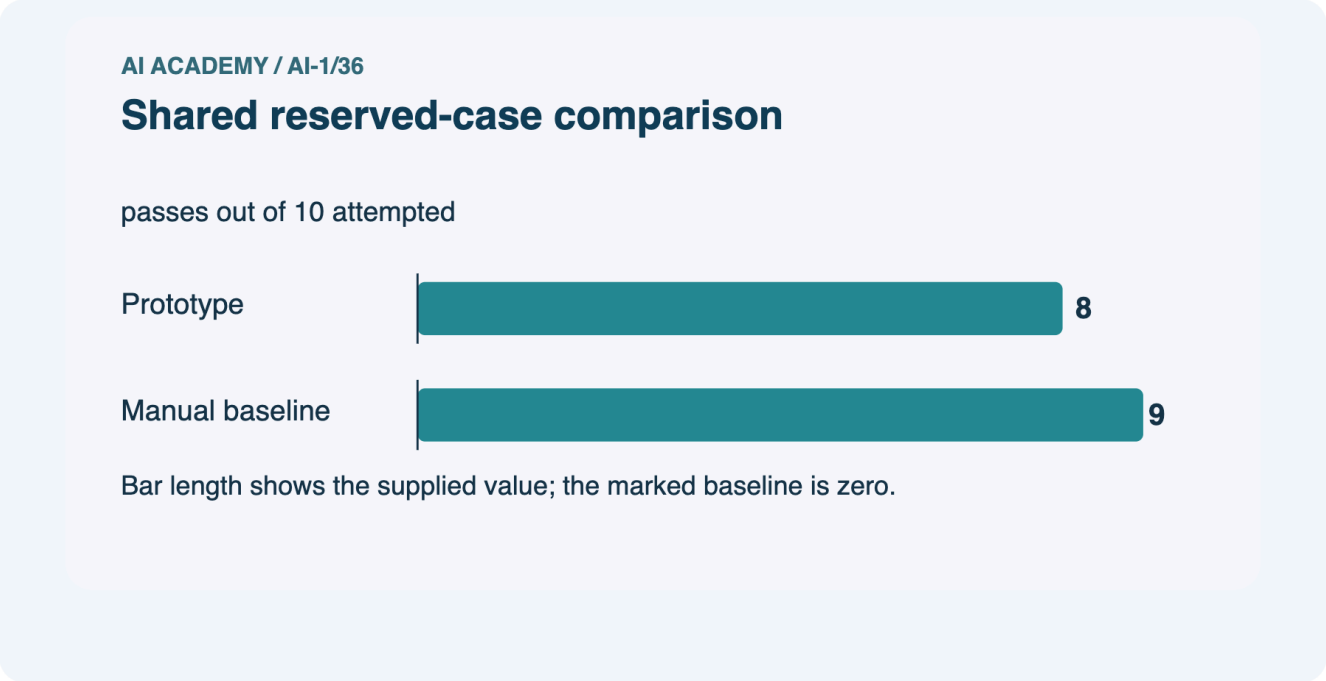


Follow the labelled picture with the matching explanation. Open the vector diagram

Describe contract: Show the task, **input**s, rule and reserved case selection. → Show results: Prototype 8/10=**80%**; **baseline** 9/10=90% on attempted cases. → Keep incomplete work: Two unrun cases remain visible, without pass/fail labels. → Defend conclusion: Prototype did not beat **baseline** on this measured outcome; explain its failures.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Shared reserved-case comparison

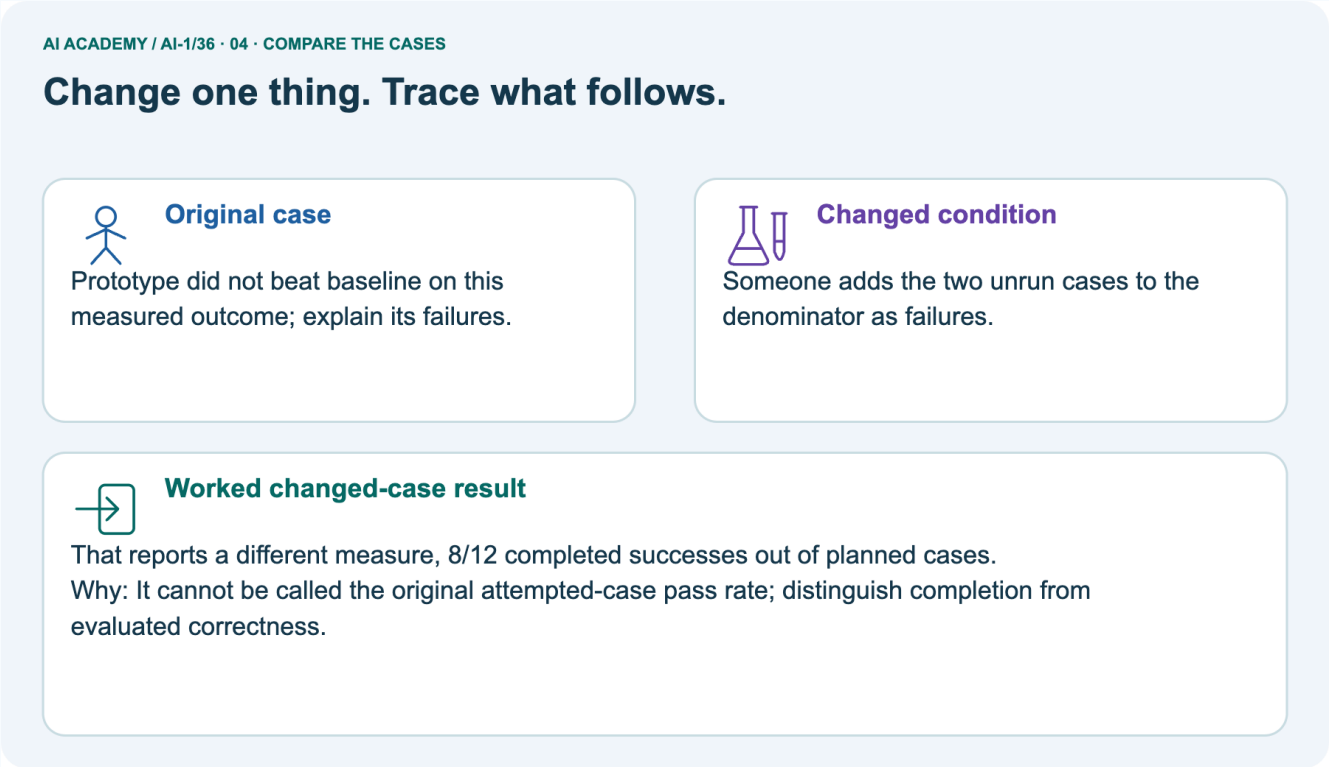


Follow the labelled picture with the matching explanation. Open the vector diagram

Shared reserved-case comparison. Units: passes out of 10 attempted. Prototype: 8. Manual **baseline**: 9.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



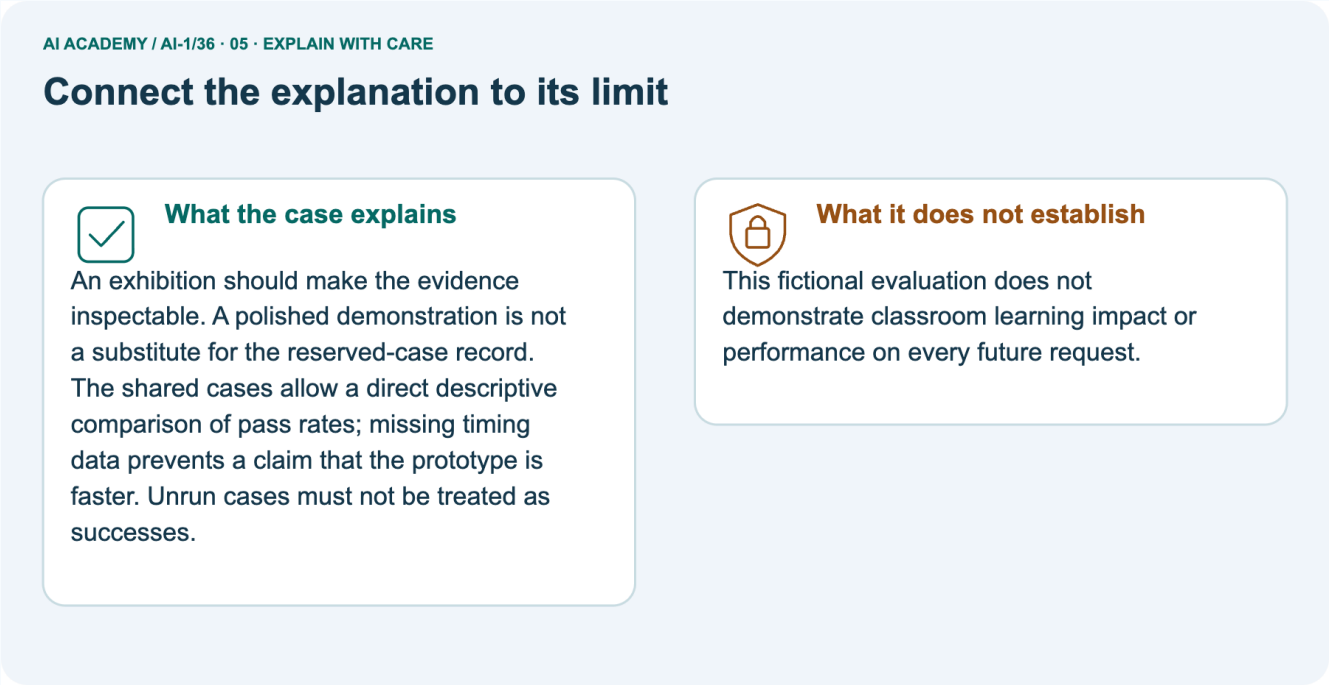
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Prototype did not beat **baseline** on this measured outcome; explain its failures.
Changed condition: Someone adds the two unrun cases to the denominator as failures.

Expected result: That reports a different measure, 8/12 completed successes out of planned cases.
Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

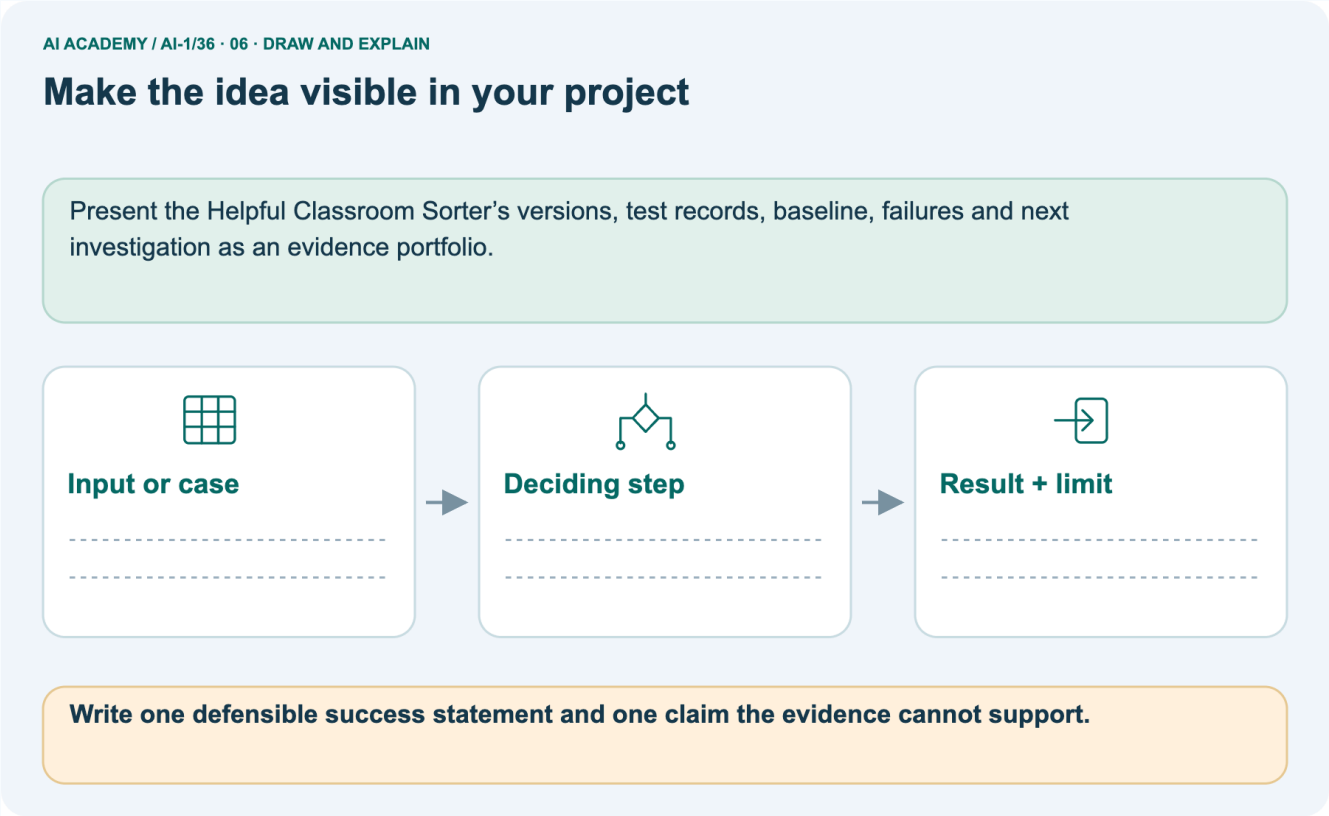
An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Limit: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio. Draw your **input**, deciding step, result and limit.

Supported question: Write one defensible success statement and one claim the **evidence** cannot support.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.